

Physics A (H156, H556)

Exploring - Using ultrasound ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is 1.
- The total number of marks may take into account some 'either/or' question choices.

- 1 A small sample of muscle has volume 1.0 cm^3 and mass 1.10 g .
The speed of ultrasound in the muscle is 1600 ms^{-1} .

What is the acoustic impedance of the muscle?

- A $1.76 \times 10^3 \text{ kg m}^{-2} \text{ s}^{-1}$
B $1.76 \times 10^4 \text{ kg m}^{-2} \text{ s}^{-1}$
C $1.76 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
D $1.76 \times 10^{12} \text{ kg m}^{-2} \text{ s}^{-1}$

Your answer

[1]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1			C	1	<u>Examiner's Comments</u> This question involves several stages, the most critical being the correct conversion of the volume units to m^3. Very encouragingly, the majority were able to do this correctly and work through to the correct response. A was a common distractor for those who made no conversion from g and cm^3.
			Total	1	